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Silicon carbide sector

Updates on AI and auto applications - AI a positive factor but how meaningful?

We provide our latest thoughts on the SiC sector, based on our industry research. We believe the data center power supply build and upcoming architecture migration from AC to HVDC (high voltage direct current) are bringing positive developments to the SiC sector. Although AI is generating strong sentiments for SiC-related stocks, we caution that expectations shall remain grounded on the contribution level. We think data centers could contribute a \$400-500mn or fewer to the SiC device TAM in 2028 or outer years. As the SiC device TAM was already built up by EV and could reach \$4.0-4.5bn in 2026, we think data center contribution could rise from current LSD to MSD to HSD within two to three years. AI will be a critical driver for the SiC sector, but its contribution level to the SiC device TAM could fall short of the bullish camp's expectations of 20-30%, in our view.

- **Data center SiC MOS - long lead time and high price.** Our industry research indicates that the lead time for AI SiC MOS has increased to 52 weeks. Additionally, prices for AI SiC MOS (high spec ones) are now at least twice as high as those for auto inverter SiC MOS, when comparing identical die size and Rds(On) specs. This encouraging dynamic is driven by exceptionally strong and urgent orders from AI customers, while qualified production capacity is temporarily limited to meet the surge in demand. Looking ahead, we think price reductions for AI SiC MOS could be much smaller than in the auto market, despite more qualified capacity.
- **Data centers potentially \$400-500mn or fewer to the SiC device TAM in 2028 or outer years.** The push towards higher rack power densities is forcing upgrades of data center power architecture from AC to HVDC, and SiC enables more efficient, compact conversion at both the infra level (UPS and SST) and the server level (PSU). Our industry research suggests 1MW HVDC data center power build could have SiC content value of \$5.0-6.0k or slightly higher, based on its power module topology and device price. J.P. Morgan teams have forecast an 80GW AI data center power new installment in 2028, implying a \$400-480mn contribution to the SiC device TAM, if HVDC penetration is 100%. Yet, the data center SiC TAM shall be smaller than this range, considering the mix between AC and HVDC.
- **SiC device TAM already built up by EV; data center contribution potentially rising to MSD to HSD in 2028 or outer years.** Although AI is generating strong sentiment for SiC-related stocks, we caution that expectations should remain grounded on the contribution level. The Yole Group forecasts ([link](#)) that the global SiC device TAM will grow from ~\$3.7bn in 2025 to ~\$4.6bn in 2026, up 25% YoY due mainly to the accelerating 800V migration in EVs. In our view, SiC is already benefiting significantly from EV adoption, and we think the global SiC device TAM could be \$4.0-4.5bn in 2026. Data center contribution to the SiC device TAM could grow from currently low single digits to mid-to-high-single digits two to three years out. We believe data centers will be a very important driver for the SiC sector, but we suspect its contribution level could fall short of the bullish camp's

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expectations of 20-30% (of the TAM).

- **SiC substrate prices - 6" stabilized but 8" dropping.** Prices for 6" SiC substrates have stabilized at ~Rmb1.5k since early 2026, vs Rmb3.0k in early 2025 and Rmb4.5k in early 2024. Some suppliers have shut down capacity given prices are below their production cash cost (Rmb1.5-2.0k, depending on scale). China's top 2 SiC substrate suppliers are operating at nearly full utilization rates for 6", but industry over-capacity prevents any meaningful price recovery. Prices for 8" SiC substrates have been in constant decline: ~Rmb7.0k in early 2025, Rmb5.5-6.0k in mid 2025, ~Rmb4.5k in early 2026 and ~Rmb4.0k in mid 2026. We expect 8" SiC substrate prices to further decline to Rmb 3.0-4.0k at end 2026 and Rmb2.5-3.5k in 2027. Customers request the pricing gap between 6" and 8" SiC substrates to be below 1.7x, based on wafer area difference. Even considering the production efficiency gain from 6" to 8", the price gap should be capped at 2.0x. Key triggers for the 8" price decline include improvements in yield and efficiency, as well as more effective supply from Chinese suppliers. The above prices are based on our industry research and refer to those offered by Chinese suppliers to large customers. A leading SiC substrate supplier talked about raising 8" prices. We believe it will be quite challenging to raise prices to major, important customers, if a supplier aims for market share. Our industry studies over the past four years and most recent industry discussions do not support the price hike expectation.
- **Auto inverter SiC MOS - ongoing price declines but increasing lead times.** Prices for auto inverter SiC MOS also continue to decline rapidly in the China market. Prices from international IDMs could fall from \$6.0-6.5 in 2024 and \$5.0-5.5 in 2025 to \$3.0-3.5 in 2026 and potentially \$2.5 in 2027. Chinese suppliers could quote 15-30% lower, depending on product competency. On the positive side, lead time for auto SiC MOS has increased from 26 weeks (the basic manufacturing time) to 39 weeks, with some specs lengthening to 52 weeks, given the AI crowd-out effect. Pricing negotiation frequency has been reduced from quarterly to semi-annual for some cases, amid the lead time increase. However, whether prices for auto SiC MOS could stabilize into 2027 is not clear, as it depends on supply-side competition. If more Chinese SiC MOS suppliers are qualified for auto inverters, prices for SiC MOS will remain under pressure into 2027.
- **Fast-rising SiC penetration in China EVs.** Ongoing SiC price declines drive SiC adoption to EV models priced ~Rmb150k. We think China's SiC EV shipments (excl. Tesla) reached 2.5mn units in 2025 and could rise to 4.0mn units or more in 2026, up 60% YoY or more. This would lift the SiC penetration rate from the mid-teens in 2025 to the low twenties in 2026. Additionally, SiC product iterations are improving on-resistance ($R_{ds(On)}$), enabling a reduction in the number of SiC MOS dies per module in some EV models (i.e., 1200V modules from 48 to 42/36 dies, and 750V modules from 36 to 30/24 dies). Net-net, we believe the auto SiC device TAM will grow moderately in 2026.
- **Implications for A-share stocks.** United Nova (UNT) is the leading Chinese supplier for SiC MOS, and it has mentioned the potential to increase SiC MOS (chip and module) revenue from Rmb1.5bn in 2025 to Rmb3.0bn in 2026 (implying a revenue mix of 18% and 28% in 2025 and 26) ([JPM note](#)). We believe UNT remains a key beneficiary of strong SiC MOS demand. UNT has sampled its 8" SiC MOS to an international AI server customer for qualification, which we believe is a CSP ASIC company. If UNT can pass qualification, we believe it would help ease the current supply tightness as UNT has mass production capabilities and UNT can ramp up more 8" SiC capacity in a timely manner. SICC is the most leading Chinese supplier for 6" and 8" SiC substrates and could benefit from increasing SiC adoption in datacenters and potentially advanced packaging in the mid to long term. SICC is ramping up its 8" SiC substrates meaningfully this year (i.e. potentially doubling YoY). However, we see continued pricing pressure for 8" SiC substrates through 2026 and we remain cautious on SICC stock, as we sense certain wrong expectations on SiC substrate pricing outlook.
- **Implications for Rohm (covered by Akinori Kanemoto).** Against FY25 revenue of just

over JPY 40bn (JPM estimate; +14% YoY, with SiC devices/modules +41% YoY), the company is planning FY26 revenue of just under JPY 55bn (JPM estimate; +30% YoY, with devices/modules +55% YoY). In FY26, management plans to offset the loss of its external 6-inch wafer sales business with growth in device/module revenue. For xEV inverter applications, China accounts for more than half of FY25 sales, but toward FY28 the China mix is expected to decline while Japan/Korea/Europe/US increase. SiC MOSFETs for AI servers have already ramped in FY25, and the company plans to lift the revenue mix to 10% in FY26. On the technology front, mass production of 8-inch wafers is scheduled to start in FY26 - initially deployed in its 4th-generation platform, followed by the rollout of 5th-generation MOS to improve yields and productivity. Including the impact of the impairment taken at FY25 year-end, the company is targeting a return to profitability by FY28.

Bookshelf – (China) SiC sector reports

- Sep 2025 - Silicon Carbide Sector - Our takes on CoWoS SiC carrier / thermal interface material; GlobalWafers at SEMICON Taiwan - [Report link](#)
- Sep 2025 - Silicon Carbide Sector - News reports on CoWoS SiC interposer; we view this as only a concept and R&D direction, without further visibility - [Report link](#)
- May 2025 - China Silicon Carbide Sector - Renesas abandons plan to produce SiC chips, with Chinese overproduction being one key reason - [Report link](#)
- Apr 2025 - China Silicon Carbide Sector - Key takeaways from our SiC substrate expert call; solid SiC EV model pipeline at the Shanghai Auto Show - [Report link](#)
- Apr 2025- China Silicon Carbide Sector - Increased competition; Updating views on substrates, device, equipment, applications and stocks - [Report link](#)
- Dec 2024 - China Tech - Geopolitical dynamics on mature node semis: US Section 301 investigation and international IDMs' local-for-local strategies - [Report link](#)
- Dec 2024 - China Silicon Carbide Sector - 2024 recap: A tough year for 6" substrates. 2025 outlook: Industry focusing on ramp of 8" substrates & auto SiC MOS; robust growth for China SiC EV - [Report link](#)
- Nov 2024 - China Silicon Carbide Sector - Key takeaways from Axcelis 3Q24 earnings call; China SiC device capacity build dynamics - [Report link](#)
- Nov 2024 - Silicon Carbide Sector - Recent industry dynamics on SiC/Si hybrid modules for EV inverters - [Report link](#)
- Sept 2024 - Silicon Carbide Sector - 8" SiC partnership of Episil Tech and VIS - [Report link](#)
- Aug 2024 - China Silicon Carbide Sector - Read-through from Silan Micro's 1H24 print and earnings call - [Report link](#)
- Jul 2024 - China Silicon Carbide Sector - China auto market landscape and substrate/device 8" transition as focal points; better demand in 2H24 - [Report link](#)
- Jun 2024 - China Silicon Carbide Sector - China auto SiC MOS: Hyper demand growth, rising competition and green shoots of localization - [Report link](#)
- Jun 2024 - United Nova - Chinese leader in auto SiC MOS; Solid revenue growth but lingering profitability pressure; Initiate at Neutral - [Report link](#)
- Mar 2024 - China Silicon Carbide Sector - China substrate competitive landscape updates from TanKeBlue and implications for SICC - [Report link](#)
- Feb 2024 - China Silicon Carbide Sector - Post-CNY checks: 'Materials' a back-end loaded 2024 and initial sign of over-supply; implications for Device Equipment - [Report link](#)
- Jan 2024 - China Silicon Carbide Sector - Chinese SiC epi maker EpiWorld IPO prospectus; implications for SICC and SiC industry - [Report link](#)
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- Jun 2023 - Silicon Carbide Sector - Our views on Chinese progress in substrates, device and equipment; OW SICC and Rohm - [Report link](#)

- Apr 2023 - Japan Electronic Components - Power semiconductor industry and position of Japanese manufacturers - [Report link](#)
- Apr 2023 - Silicon Carbide Sector - Asian SiC foundries riding on strong market demand and localization benefits; HANA & Episil Tech - [Report link](#)
- Feb 2023 - First Principles - Silicon Carbide: Key growth driver for European semi device makers - [Report link](#)
- Feb 2023 - Networking & Hardware: The Silicon Carbide (SiC) Cheat Sheet - [Report link](#)
- Feb 2023 - Silicon Carbide Deep Dive: Substrate and Asia focus; positive about the SiC sector and OW WOLF, COHR, Rohm and SICC - [Report link](#)
- Feb 2023 - SICC - Initiate pure China SiC substrate play at OW, on TAM expansion, localization and power SiC ramp - [Report link](#)

Companies Discussed in This Report (all prices in this report as of market close on 25 May 2026, unless otherwise indicated)
Rohm (6963)(6963.T/¥5,030/OW), United Nova - A(688469.SS/Rmb8.25/N)

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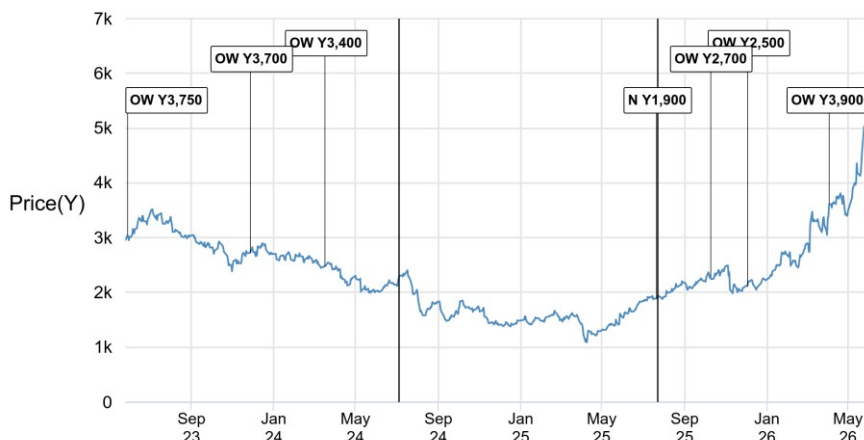
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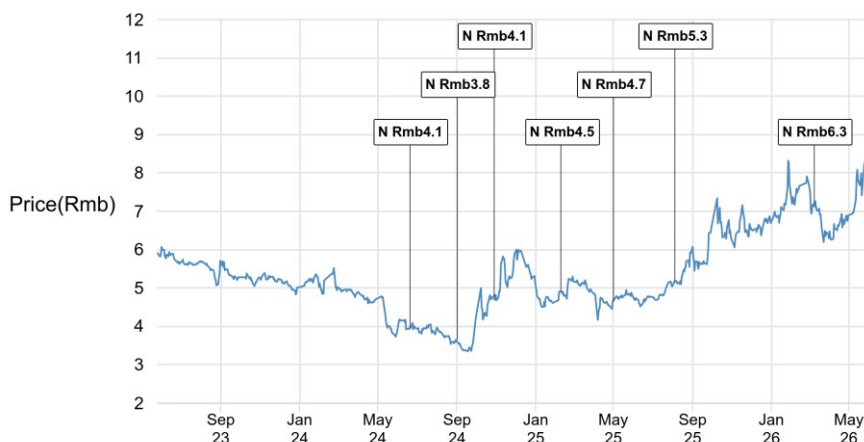
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Date	Rating	Price (Y)	Price Target (Y)
30-May-23	OW	3042	3,750
28-Nov-23	OW	2725	3,700
18-Mar-24	OW	2472	3,400
22-Jul-25	N	1888	1,900
10-Oct-25	OW	2294	2,700
04-Dec-25	OW	2116	2,500
04-Apr-26	OW	3608	3,900

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Aug 12, 2001. All share prices are as of market close on the previous business day.
Break in coverage Jul 04, 2024 - Jul 22, 2025.

United Nova - A (688469.SS, 688469 CH) Price Chart



Date	Rating	Price (Rmb)	Price Target (Rmb)
21-Jun-24	N	3.98	4.1
02-Sep-24	N	3.65	3.8
29-Oct-24	N	4.79	4.1
10-Feb-25	N	4.91	4.5
02-May-25	N	4.64	4.7
05-Aug-25	N	5.18	5.3
09-Mar-26	N	7.18	6.3

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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